

Loren Deklerk

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Portfolio: <https://deklerk.netlify.app> | GitHub: <https://github.com/Biden254> | LinkedIn: <https://www.linkedin.com/in/loren-deklerk>

Professional Summary

Cloud-focused Computer Science student with hands-on experience in **Linux-based systems, networking fundamentals, AWS cloud services, and Python scripting**. Proven ability to **troubleshoot distributed systems, analyze root causes, and build tools that improve operational efficiency**. Experienced in working with APIs, databases, and real-time systems, with strong foundations in **TCP/IP, DNS, HTTP, OS concepts, and cloud architecture**. Passionate about customer-facing technical support, automation, and continuous learning in high-impact cloud environments.

Technical Skills

Programming & Scripting

- Python, C, C++, Java, JavaScript
- Shell scripting (Linux), basic automation scripts
- Data Structures & Algorithms

Cloud & Systems

- AWS (EC2, IAM, S3, Cloud fundamentals – via AWS re/Start)
- Linux/Unix environments (CLI, process management, logs)
- Windows OS fundamentals

Networking Fundamentals

- OSI Model, TCP/IP, HTTP/HTTPS, DNS, DHCP
- CIDR notation, subnets, NAT, routing & switching basics
- Client-server architectures

Databases & Storage

- Relational database concepts
- Data modeling, structured data handling
- Storage fundamentals & data pipelines

Software Development & Support

- Troubleshooting & root cause analysis
- API integration (REST, JSON)
- Object-Oriented Programming (OOP)
- Git/GitHub, Agile/Scrum workflows
- Documentation & technical communication

Automation & Tools

- Python-based tools and scripts
- Log analysis and debugging
- Process improvement through tooling

Education

Bachelor of Computer Science (B.C.S.)

Dedan Kimathi University of Technology | 2024 – Ongoing

Expected Graduation: 2028 (on track for 2:2)

Experience

Software Engineering Intern (Virtual)

Electronic Arts (EA) — *November 2025*

- Supported a **production-scale C++ codebase**, troubleshooting issues and implementing fixes to improve system reliability and performance.

- Analyzed existing logic and **identified root causes** of defects, applying optimized data structures to enhance efficiency.
 - Designed and documented **object-oriented system components**, improving maintainability and knowledge transfer across teams.
 - Collaborated with global engineers using **Git-based workflows and Agile practices**, ensuring clear communication and traceability of changes.
 - Strengthened experience working in **complex, distributed software environments** similar to cloud-based systems.
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Projects

AI Dispatch System — Real-Time Systems & Troubleshooting

Python | Linux | Git

GitHub: <https://github.com/Biden254/AI-Dispatch-System>

- Built a real-time dispatch simulation system using Python coroutines, simulating cloudstyle event handling and concurrent workloads.
 - Designed a data pipeline to ingest, prioritize, and process events, mirroring production support scenarios involving high-throughput systems.
 - Applied logging, structured data handling, and debugging techniques to trace system behavior and diagnose failures.
 - Improved simulated response efficiency by 25% through algorithm optimization and resource allocation tuning.
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EduAI — Distributed Application Architecture

Python | APIs | System Design

GitHub: <https://github.com/Biden254/Edu-AI>

- Architected a multi-component application with clear separation of services, similar to cloud-native design patterns.
 - Refactored inefficient logic to reduce memory usage by 30%, demonstrating performance tuning and system optimization skills.
 - Designed API contracts and data flows, supporting scalable integration between components.
 - Documented system behavior and workflows to support maintainability and onboarding.
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GenAI Web Application — API Integration & Debugging

Python | JavaScript | REST APIs

GitHub: <https://github.com/Biden254/GEN-AI>

- Developed a web application integrating **external APIs**, handling JSON requests/responses and error scenarios.
 - Implemented **robust error handling and validation**, ensuring reliability in client-server communication.
 - Translated complex backend processes into user-facing workflows, strengthening customer-centric development skills.
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Certifications & Training

- AWS re/Start Program — Amazon Web Services, 2025
 - Cisco Python Certification — Cisco, 2024
 - Cisco Networking Devices & Initial Configuration — Cisco, 2024
 - Software Engineering Bootcamp — Power Learn Project (PLP), 2025
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Extracurricular Activities

AWS Club

- Organized and participated in cloud computing workshops, helping peers understand AWS concepts and cloud fundamentals.
- Strengthened technical communication skills by explaining networking, cloud, and Linux concepts to non-experts.

Cybersecurity Society

- Participated in CTF challenges and secure coding discussions, improving systematic troubleshooting, attention to detail, and risk awareness.